

Student's assessment number



UNITED REPUBLIC OF TANZANIA
PRIME MINISTER'S OFFICE
WANGING'OMBE DISTRICT COUNCIL



FORM TWO SERIES ONE PRE- NECTA EXAMINATION

PHYSICS

Duration: 2 :30 Hours

SEPTEMBER 2026

Instructions

- 1.This paper consists of sections A, B and C with a total of 10 questions
- 2.Answer all questions in the spaces provided
- 3.Section A and C carries fifteen (15) marks each and section B carries seventy (70) marks
4. Calculators and mathematical tables may be used
- 5.The following information may be useful
 - i) Acceleration due to gravity, $g=10\text{m/s}^2$
 - ii) Pie $\pi = 3.14$
 - iii) Density of water = 1000kg/m^3 or 1g/cm^3 .

FOR EXAMINOR'S USE ONLY

QUESTION NUMBER	SCORE	EXAMINOR'S INITIALS
1		
2		
3		

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9		
10		
TOTAL		

SECTION A (15 Marks)

Answer all questions in this section.

1. For each of the following items (i-x), choose the correct answer from the given alternatives and write its letter in the box provided
 - i) The branch of physics that deals with the interaction between electric and magnetic fields are
 - A. Electronics
 - B. Magnetic
 - C. Electric current
 - D. Electromagnetism
 - ii) Which of the following is the fundamental unit?
 - A) Kilogram
 - B) Newton
 - C) Joule
 - D) Mass
 - iii) A mass of one kilogram is pushed by the gravitational force towards the centre of the earth by force of
 - A) 1 newton
 - B) 10 newton
 - C) 100 newton
 - D) 1000newtons
 - iv) A substance has a mass of 4000g and volume of 100cm³. What is its density in kg/m³?
 - A) 0.25kg³
 - B) 250kg/m³
 - C) 4kg/m³
 - D) 4000kg/m³
 - v) Why does a stone sink in water?
 - A) Because its volume is large
 - B) Because its mass is small
 - C) Because its weight is greater than the up thrust
 - D) Because its density is less than that of water
 - vi) When charging a body by rubbing with either fur or silk the particles which are transformed are:
 - A) protons and electrons
 - B) Protons

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- C) Electrons
- D) neutrons
- vii) Electric current is expressed as
 - A) Quantity of charge per unit time
 - B) Voltage per unit time
 - C) Resistance per unit voltage
 - D) Charge per unit voltage
- viii) A glassy capillary tube is placed in a beaker of water. the water level inside the tube is higher than the water level outside due to:
 - A) Cohesion
 - B) Adhesion
 - C) Surface tension
 - D) Elasticity
- ix) which devices can be used to demonstrate the existence of atmospheric pressure?
 - A) Siphon and plunger
 - B) Magdeburg hemisphere and force pump
 - C) Magdeburg hemisphere and crushing can
 - D) Crushing can and syringe
- x) Work done and energy are related in that
 - A) They are both a result of mechanical process
 - B) They have the same unit in which they are measured
 - C) They all explain how fast a certain task is performed
 - D) All are typical examples of chemical energy.

2. Match the functions of instruments in list A with their corresponding instruments in list B by the writing the correct answer bellows the item number in the table provided

List A	List B
i) measure the atmospheric pressure	A. siphon
ii) measure the gas pressure in car tires	B. Lift pump
iii) used to free blocked water sink	C. Syringe
iv) used for sucking and ejecting liquid by increasing and decreasing pressure	D. Vacuum cleaner
v) allow water to flow because of pressure differences	E. barometer
	F. plunger
	G. Pressure gauge

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List A	i	ii	iii	iv	v
List B					

SECTION B

3. Sharp edges of knife cut easily than blunt knife when the same force is applied

a) Use the pressure formula to explain this observation (4 marks)



b) Explain what would happen if the force applied to the blunt knife is increased significantly. (6 marks)



4. a) Explain why a slight electrical shock is experienced by someone who rub feet on a carpet and touch a metallic object. (4 Marks)



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b) In an experiment to demonstrate the existence of electrostatic force, a form two student rubbed and comb through his hair and brought it near the pieces of paper for about 15s.

i) State what happened when the comb was brought near the pieces of paper (3 marks)

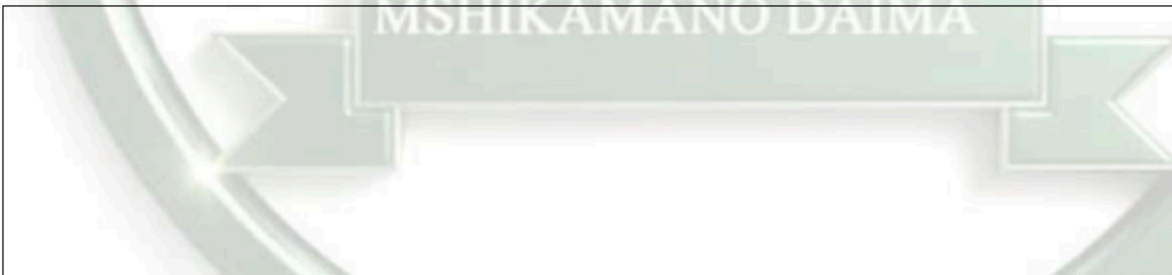


ii) Explain what happened o the pieces of paper on the comb after few seconds (3 marks)



5. a) A car starting form rest has a uniform acceleration of 0.75m/S^2

How long will it take to acquire a velocity of 54km/h ? (4 marks)



b) whenever the velocity (magnitute or direction or both) of an object changes, we say that the object is accelerative. Under what circumstance is an object not accelerated (6marks)

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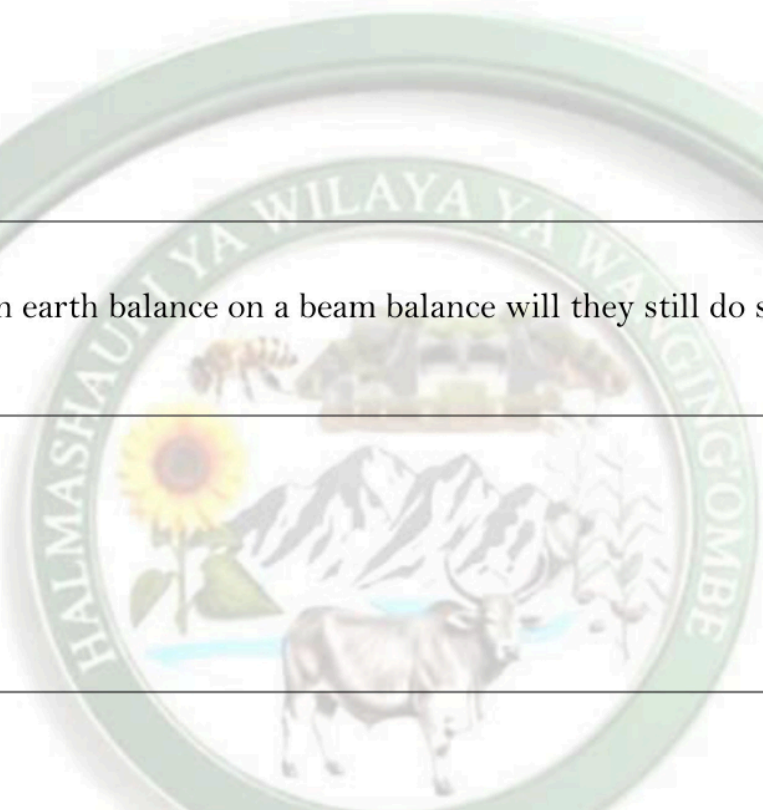
6.A) A hydrometer is an instrument used to measure densities of liquids. Explain why the bulb of hydrometer is weighed with mercury or lead shots? (4 marks)

b) The mass of a substance is 40g. The mass of an equal volume of water is 20g. Calculate the relative density of a substance (6 Marks)

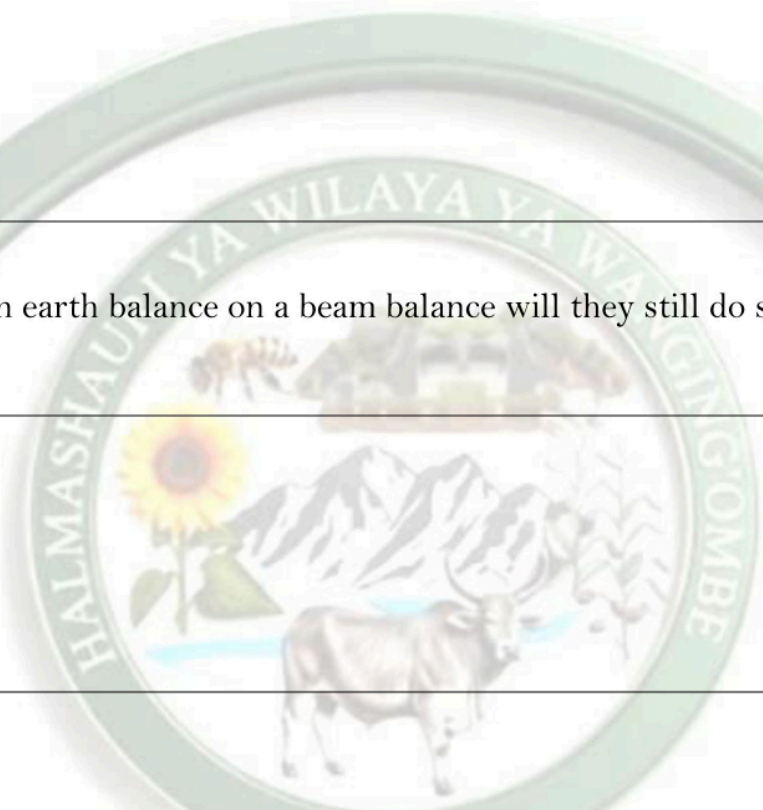
7. a) Outline the two conditions for work to be done (4 marks)

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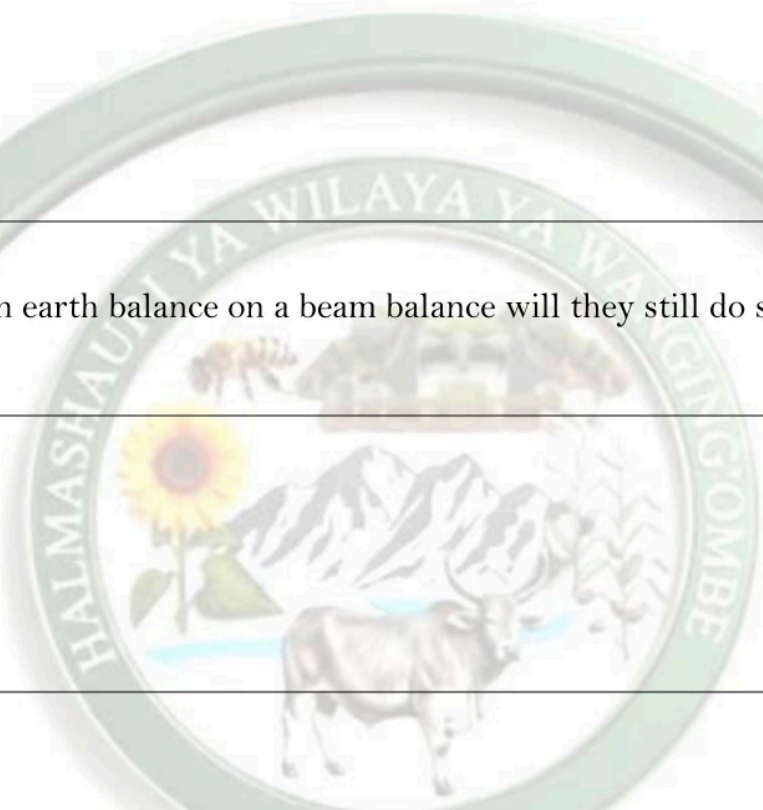
b) An escalator carries 100 people of average mass 70kg to a height of 6m in minute. Find the power necessary to do this. (6 Marks)



8. a) Two objects on earth balance on a beam balance will they still do so on the moon? Why?
4 Marks



b) A beaker contains 100cm^3 of the liquid x. A 20cm^3 pipette is used four times to transfer the liquid to another beaker. If the mass of the remaining liquid in the original beaker is 20g, find the density of the liquid x and hence identify liquid x. 6(Marks)



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9. a) A spring balance has a scale – pan of 0.4N weight etched to it. A load of 2.0N placed on the pan exert the spring 24mm from its origin un stretched length. Calculate the load on the scale pan when the extension is 16mm. (Assume the elastic limit is not exceeded)



b) Amina and Mwajuma were drinking soup. The soup of Amina was hot while the soup of Mwajuma was cold.

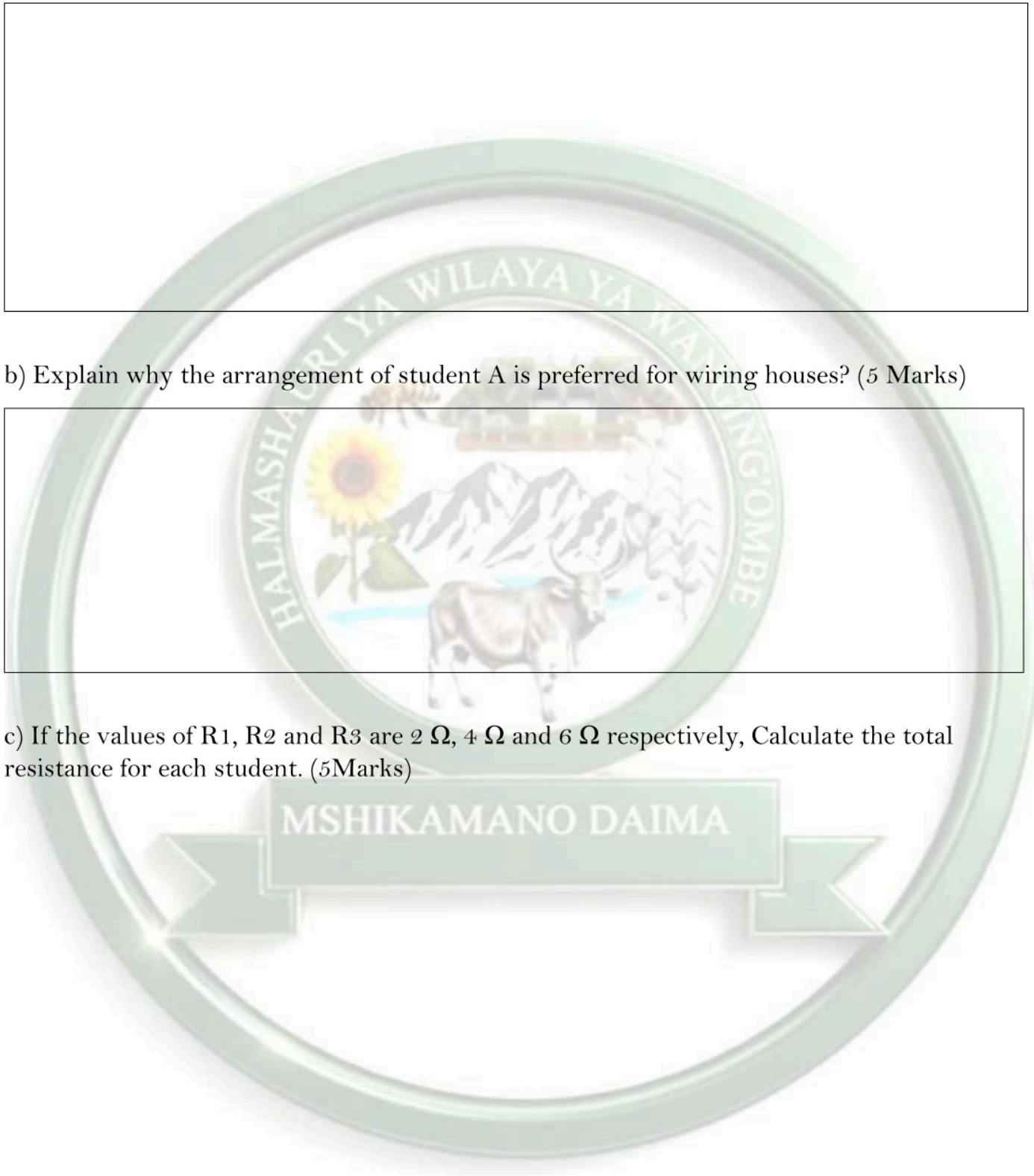
On exchanging their soup bowls, Mwajuma wondered that Amin's soup was testier than her soup and Amina agreed with Mwajuma. As a physics students explain why Amina's soup was testier than that of Mwajuma (4Marks)



10. Two students designed circuit using three resistors labeled R_1 , R_2 and R_3 connected to an electric source labeled V . Student A connected them in parallel and student B connected them in series.

a) Show that the total resistance for student B is given as $R_T = R_1 + R_2 + R_3$ (5 Marks)

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b) Explain why the arrangement of student A is preferred for wiring houses? (5 Marks)

c) If the values of R_1 , R_2 and R_3 are $2\ \Omega$, $4\ \Omega$ and $6\ \Omega$ respectively, Calculate the total resistance for each student. (5Marks)

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FORM TWO MID-TERM EXAMINATION, SEPTEMBER 2023
PHYSICS - MARKING SCHEME

01.

i	ii	iii	iv	v	vi	vii	viii	ix	x
D	A	B	D	C	C	A	B	B	B

02.

LIT A	i	ii	iii	iv	v
LIT B	E	G	F	C	A

03. a) Pressure formula is expressed as $\text{Pressure} = \text{Force} / \text{Area}$.

From the formula, Pressure increases as the area decreases provided that force is constant. Hence sharp knife cuts easily since it has smaller area than blunt knife. (04 marks)

b) Applying more force to a blunt knife increases the risk of the blade slipping and causing injury. (02 marks)

04. a) This happens because rubbing feet on carpet builds up static electric charge by transferring electrons between surfaces. When touching a metallic object, that charge rapidly equalizes and a small electric shock is experienced. (04 marks)

b) i) The comb picked up pieces of paper (02 marks).

ii) The papers dropped off ~~(02 marks)~~ because they acquire similar charges as the comb. (03 marks)

05. a)

Soln.

Data given

Initial velocity (u) = 0 m/s

Acceleration (a) = 0.75 m/s^2

Final velocity (v) = 54 km/h

Time (t) = ?

(07 1/2 marks)

100-TERM EXAM SCHEME

From: $V = ut + at$. (0.5 marks)

$$54 \text{ km/h} = 0 \text{ m/s} + 0.75 \text{ m/s}^2 \times t$$

$$54 \text{ km/h} = 0.75 \text{ m/s}^2 \times t$$

$$t = \frac{54 \text{ km/h}}{0.75 \text{ m/s}^2}$$

$$\text{but } 54 \text{ km/h} = \frac{54 \times 1000 \text{ m}}{3600 \text{ s}}$$

$$= 15 \text{ m/s. (0 marks)}$$

$$\text{Thus: } t = \frac{15 \text{ m/s}}{0.75 \text{ m/s}^2}$$

$$= 20 \text{ seconds. (02 marks)}$$

5b. An object is not accelerating if it is moving at a constant speed in a straight line (06 marks).

6a. To keep it upright when floating in a liquid (04 marks).

b,

Soln

Data given

Mass of cube stone (m_0) = 40g

Mass of an equal volume of water (m_1) = 20g (0.5 marks)

Relative density (RD) = ?

$$\text{from: } RD = \frac{m_0}{m_1} \text{ (0.5 marks)}$$

$$= \frac{40 \text{ g}}{20 \text{ g}} \text{ (01 mark)}$$

$$= 2, \text{ (03 marks)}$$

- 7.9) i) There must be a force acting on an object (0.2 marks)
 ii) The object must move along the direction of the applied force (0.2 marks).

b,

Soln.

Data given.

Mass of one person (m) = 70 kg

Distance (d) = 6 m

Time (t) = 1 minute.

Power (P) = ?

From: $\text{Power} = \frac{\text{Work done}}{\text{time taken}}$ (0.5)

$$= \frac{\text{Force} \times \text{distance}}{\text{time}} \quad (0.5)$$

$$= \frac{\text{mass} \times \text{gravitational pull} \times \text{distance}}{\text{time}} \quad (0.5)$$

$$= \frac{(70 \times 100) \times 10 \times 6}{60} \quad (0.1)$$

$$= \underline{7000 \text{ W}} \quad (0.3 \text{ marks}).$$

8 a) Yes, and this is because mass do not vary, it is always constant. (04 marks).

b)

Soln.

Data given.

Initial volume in the beaker (V_0) = 100 cm^3

Volume taken out of the beaker (V_t) = $4 \times 90 \text{ cm}^3$ (vol₂)

Mass of the remaining liquid (m_L) = 20 g

Density of the liquid (ρ_L) = ?

$$\text{from } \rho_L = \frac{m_L}{V_r}$$

Volume of the remaining liquid (V_r) (vol₂)

$$= \frac{20 \text{ g}}{V_r} \quad \text{--- i)}$$

$$\begin{aligned} \text{but } V_r &= V_0 - V_t \\ &= 100 \text{ cm}^3 - 80 \text{ cm}^3 \\ &= 20 \text{ cm}^3 \quad \text{--- ii)} \end{aligned}$$

substituting eqn ii) into eqn i).

$$\begin{aligned} \rho_L &= \frac{20 \text{ g}}{20 \text{ cm}^3} \quad (01 \text{ mark}). \\ &= \underline{1 \text{ g/cm}^3} \quad (02 \text{ marks}). \end{aligned}$$

Liquid X is water since its density is 1 g/cm^3 (02 marks).

09 a)

Soln.

Data given.

Weight of the scale pan (W_0) = 0.4 N

Weight added to the pan (W_1) = 2.0 N

(00/2 marks)

Weight added to the pan (W_2) = ?

Extension produced by $W_0 + W_1$, $e_1 = 24$ mm

Extension produced by $W_0 + W_2$, $e_2 = 16$ mm

from Hooke's law,

$$F \propto e$$

$$F = k e \quad (00/2 \text{ marks})$$

$$\frac{F}{e} = k$$

$$\frac{F_1}{e_1} = \frac{F_2}{e_2}$$

$$\frac{F_1}{F_2} = \frac{e_1}{e_2}$$

$$\text{but } F_1 = W_0 + W_1$$

$$\frac{W_0 + W_1}{F_2} = \frac{e_1}{e_2}$$

$$\frac{0.4 \text{ N} + 2.0 \text{ N}}{F_2} = \frac{24 \text{ mm}}{16 \text{ mm}}$$

$$\frac{2.4 \text{ N}}{F_2} = \frac{24}{16} \quad (01 \text{ mark})$$

$$\frac{2.4 \text{ N}}{F_2} = \frac{3}{2}$$

$$3 F_2 = 2.4 \text{ N} \times 2$$

$$F_2 = \frac{2.4 \text{ N} \times 2}{3}$$

$$F_2 = 1.6 \text{ N} \quad (02 \text{ marks})$$

but $F_2 = W_1 + W_2$

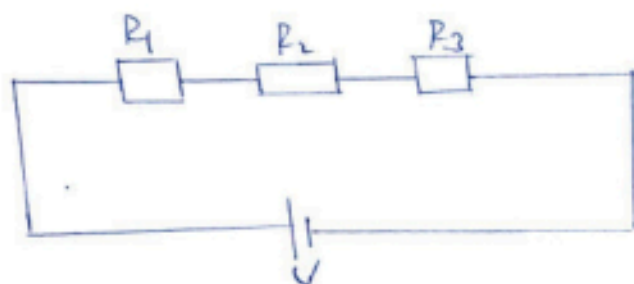
$$1.6\text{N} = 0.4\text{N} + W_2$$

$$W_2 = 1.6\text{N} - 0.4\text{N}$$

$$= \underline{1.2\text{N}} \quad (0.2 \text{ marks})$$

b) Amine's soup was tastier due to lower surface tension since it was hot. (0.4 marks)

10 a) Circuit diagram for student B is as shown below



0.1 marks

For series arrangement, p.d across, $R_1 = V_1$, $R_2 = V_2$ & $R_3 = V_3$
and the current through R_1 , R_2 & R_3 is the same, that is
Thus:

$$V = V_1 + V_2 + V_3 \quad (0.1 \text{ mark})$$

$$IR_T = IR_1 + IR_2 + IR_3 \quad (0.1 \text{ mark})$$

Dividing by I throughout:

$$\boxed{R_T = R_1 + R_2 + R_3} \text{ Hence shown. } (0.2 \text{ marks})$$

b) The arrangement of student A (parallel arrangement) is preferred because it provides independent operation for each device. If one part breaks or turns off, the rest of the circuit keeps working normally. (2.5 marks).

c)

Soln.

Data given

$$\text{Resistance } (R_1) = 2\Omega$$

$$\text{Resistance } (R_2) = 4\Omega$$

$$\text{Resistance } (R_3) = 6\Omega$$

$$\text{Resistance } (R_T) = ?$$

Case 1: For student A.

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \quad (01)$$

$$\frac{1}{R_T} = \frac{1}{2} + \frac{1}{4} + \frac{1}{6} \quad (01)$$

$$\frac{1}{R_T} = \frac{6+3+2}{12}$$

$$\frac{1}{R_T} = \frac{11}{12}$$

$$11 R_T = 12$$

$$R_T = \frac{12}{11} \Omega$$

$$R_T = 1.1\Omega \quad (0\frac{1}{2})$$

Case 2: For student B,

$$R_T = R_1 + R_2 + R_3 \quad (01)$$

$$= 2\Omega + 4\Omega + 6\Omega \quad (01)$$

$$= 12\Omega \quad (0\frac{1}{2})$$